Can We Forecast the Crude Oil Futures Spread? A Comparison of ARIMA and ARIMAX Models

Financial Econometrics - Group 7

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Executive Summary

This study evaluates whether econometric models can forecast the WTI crude oil futures spread and whether such forecasts can generate economically meaningful trading strategies. Using monthly data, we compare a baseline ARIMA model with an ARIMAX specification that incorporates macroeconomic and market-based variables. Forecast performance is assessed using both non-rolling and rolling window out-of-sample frameworks.

Across all evaluations, the ARIMA model consistently outperforms ARIMAX in terms of forecast accuracy, directional hit rate, and economic performance. While ARIMAX improves certain in-sample diagnostics, these gains do not translate into superior out-of-sample results, particularly during periods of heightened volatility such as the COVID-19 crisis. Rolling window forecasts show that ARIMA delivers more stable and robust predictions, which translate into higher cumulative returns, stronger risk-adjusted performance, and lower drawdowns in a trading context.

The underperformance of ARIMAX is partly driven by data limitations, as aligning exogenous variables substantially reduces the effective sample size and increases estimation uncertainty. Given these constraints, more complex machine learning models are not considered appropriate. Overall, the results highlight that parsimonious, price-based models such as ARIMA can provide robust and tradeable forecasts in oil futures markets.

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1 Motivation

Crude oil futures markets play a central role in global energy pricing, risk management, and speculative trading. For market participants, the ability to form a reliable forward-looking view on oil price dynamics, particularly across the futures curve, can generate substantial economic value. While short-horizon movements in financial asset prices such as equities are often well approximated by a random walk, oil futures markets exhibit structural features that may allow for a degree of predictability.

In particular, the spread between near-dated futures contracts reflects underlying economic forces such as inventory dynamics, storage costs, convenience yields (cost of carry), and expectations about future supply and demand conditions. Unlike spot prices, which are heavily influenced by contemporaneous shocks, futures spreads embed information about market tightness and are therefore less likely to follow a pure random walk. Moreover, the oil market is known to display seasonal patterns and mean-reverting behaviour, driven by factors such as cyclical demand, production adjustments, and inventory management. These characteristics suggest that econometric forecasting techniques may be well suited to capturing systematic variation in oil futures spreads.

The objective of this project is to assess whether econometric models can be used to enhance the forward outlook on oil futures spreads and generate forecasts with both statistical and economic relevance. By exploiting historical price information and incorporating economically meaningful predictors, this study evaluates whether forecasts can outperform simple benchmark models and deliver improved out-of-sample performance. Beyond statistical accuracy, the analysis also considers whether such forecasts can be translated into a trading strategy that produces economically meaningful returns.

Overall, this project aims to bridge econometric forecasting techniques with practical decision-making in energy markets, providing insights that are relevant not only from an academic perspective but also for practitioners seeking to extract value from predictable structure in oil futures prices.

2 Main Objective and Research Questions

The main objective of the study is to develop and compare two time-series models to forecast the WTI oil future spread (CL2-CL1) and evaluate their out-of-sample (OOS) predictive accuracy and assess their practical applicability for oil trading strategies:

1. Which modelling approach provides the most accurate forecasts of oil futures spread?
2. Do macroeconomic and market variables improve forecasting performance beyond time-series dynamics?
3. Can forecast accuracy be translated into genuine efficient trading strategies?

3 Data & Methodology

3.1 Dataset Overview & Determining Time Frequency

The dataset we use contains crude oil futures prices alongside various macroeconomic indicators and oil-related stock price variables, all observed at a monthly frequency. The primary variables of interest are the front-month (CL1) and second-month (CL2) crude oil futures contracts, from which we can construct the oil futures spread.

The choice of forecasting frequency is guided by both economic considerations and data availability. Forecasting at a daily horizon is unlikely to be informative as high-frequency oil price movements are highly volatile and often well approximated by a random-walk. Whilst a weekly horizon can mitigate this noise and be a potential candidate, the majority of explanatory variables are recorded at a monthly frequency, hence making consistent weekly modelling impractical.

Therefore, this paper adopts a monthly forecasting framework, which aligns naturally with the available data and allows the models to exploit slower-moving dynamics such as seasonality and mean reversion in oil futures spreads. By combining futures price information with macroeconomic and oil-market variables within a unified econometric framework, the analysis aims to produce a more holistic and economically grounded forecast with improved statistical and practical relevance.

3.2 Target Variable Selection

The primary dependent variable of interest is the spread, which is constructed as the difference between the first and second month of WTI crude oil future contracts.

$$\text{Log Oil Spread} = ln(\text{CL}_2) - ln(\text{CL}_1)$$

The choice was economically motivated by the distinctive forecasting properties of spreads relative to price levels. The spread serves as a valuable indicator of market conditions, signalling either constant (CL2 $>$ CL1), indicating excess supply or storage availability) or backwardation (CL2 $<$ CL1), reflecting supply shortages or strong near-term demand.

Moreover, a logarithmic transformation is applied to stabilise variance and reduce the influence of large, transitory fluctuations. Formal unit root and stationarity tests (ADF and KPSS) provide mixed evidence for the stationarity of the level spread, motivating the use of a log-transformed specification that more closely satisfies weak stationarity conditions. Modelling log spreads also allows changes in the futures curve to be interpreted in relative terms, improving comparability across time and price regimes. As a result, the log-transformed spread provides a more robust target variable for econometric forecasting and out-of-sample performance evaluation.

3.3 Determining Sample Split

The dataset was split using a 70:30 (training:testing) ratio to evaluate out-of-sample forecasting performance. Unlike random splitting common in cross-sectional studies, this split preserves the time-series structure and prevent look-ahead bias by ensuring that all training observations precede all test observations chronologically. Models are estimated on the earlier portion of the sample and evaluated on the later partition, mimicking a real-world scenario where traders make forecasts using only historical information to predict future market conditions.

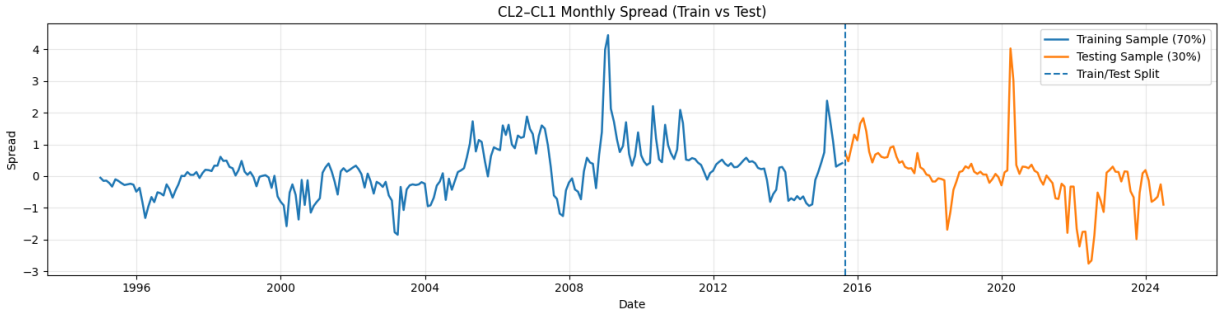
3.4 Exploratory Data Analysis

3.4.1 Visualising Sample Split

Following the data split, Figure 1 illustrates the chronological training and testing split used for out-of-sample evaluation. The split is implemented strictly in time, ensuring that model estimation relies solely on information available prior to the forecast period and avoiding look-ahead bias.

Additionally, the training sample spans multiple market environments, including periods of elevated volatility and macroeconomic stress, ensuring that model estimation is not driven by a single financial regime. Importantly, the training period excludes the COVID-19 episode, allowing the out-of-sample evaluation to assess model performance under a previously unseen and highly disruptive market environment.

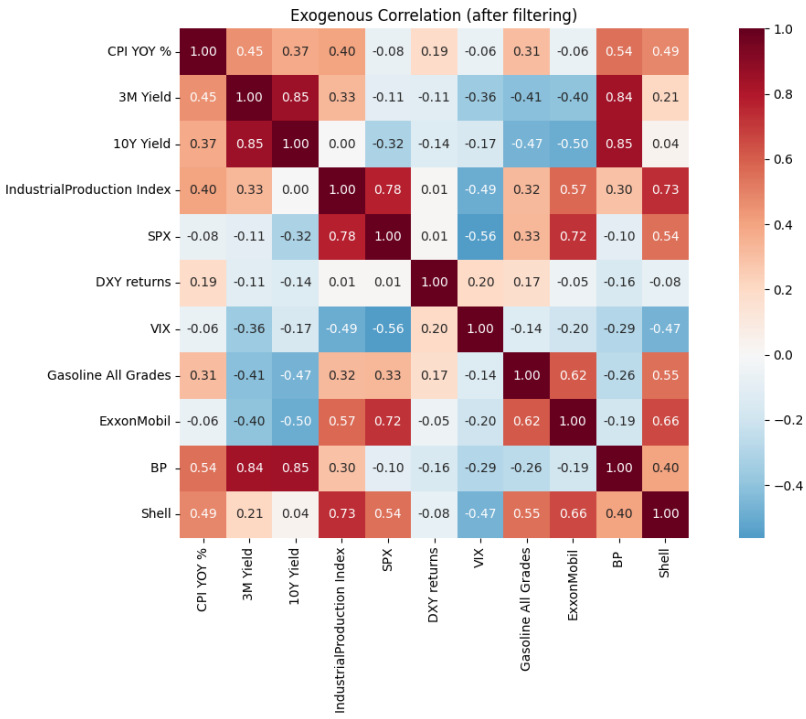
Figure 1: Training/Testing Set Splits of ln(CL2-CL1) Spread



3.4.2 Assessment of Exogenous Variables

Figure 2 provides an exploratory overview of potential exogenous variables. Correlation analysis is used solely as a preliminary screening tool to exclude near-duplicate information, while formal variable selection and multicollinearity diagnostics are conducted during ARIMAX model estimation. For further multicollinearity control, an iterative Variance Inflation Factor (VIF) procedure was applied, removing variables with VIF greater than 10.

Figure 2: Heatmap of Potential Exogenous Variables



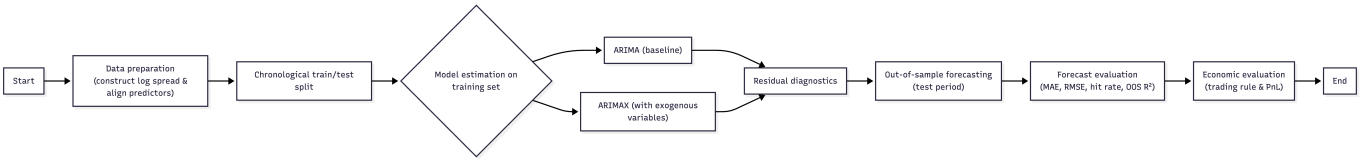
3.5 Methodology

The paper follows a structured econometric forecasting framework to evaluate the predictive performance of ARIMA and ARIMAX models for oil futures spreads. After constructing the log oil futures spread and aligning all explanatory variables at a monthly frequency, the dataset is split chronologically into training and testing samples to avoid look-ahead bias.

Model estimation is conducted exclusively on the training set, with an ARIMA model serving as a baseline and an ARIMAX specification incorporating selected macroeconomic and oil-market variables. Model adequacy is assessed using residual diagnostics to ensure that serial dependence is sufficiently captured.

Forecasts are then generated over the out-of-sample test period and evaluated using standard forecast accuracy measures, including MAE, RMSE, hit rate, and out-of-sample R^2 relative to benchmark models. Finally, forecast signals are translated into a trading strategy to assess the economic relevance of the models through cumulative returns and profit-and-loss analysis. An overview of the process can be seen in Figure 3:

Figure 3: Overview of Training and Forecasting Process



Refer to Figure 12 for full flowchart

4 Model Selection & Estimation

4.1 Stationarity Analysis

Stationarity of the spread series was assessed using the Augmented Dickey-Fuller (ADF) and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests, which evaluate complementary null hypotheses of non-stationarity and stationarity, respectively.

ADF (Augmented–Dickey Fuller) Test

$$H_0 : \delta = 0 \text{ (Suggests non-stationarity)} \quad H_1 : \delta < 0 \text{ (Suggests stationarity)}$$

KPSS (Kwiatkowski–Phillips–Schmidt–Shin) Test

$$H_0 : \delta = 0 \text{ (Suggests stationarity)} \quad H_1 : \delta < 0 \text{ (Suggests non-stationarity)}$$

The ADF test strongly rejects the unit root hypothesis ($p < 0.00001$), indicating stationarity. However, the KPSS test rejects stationarity at the 5% level ($p = 0.0230$), producing conflicting results (shown in Table 5). The conflicting evidence suggests that the series is highly persistent and potentially near integrated. To resolve the ambiguity, ARIMA models were estimated with both d=0 and d=1, with final selection based on information criteria. In addition, the ACF shows slow and gradual decay, while the PACF exhibits a sharp spike at lag 1. Therefore, the following pattern is consistent with a low-order autoaggressive structure.

4.2 ARIMA Model (Baseline)

Model selection was conducted using an automated search over ARIMA specifications with autoregressive and moving average orders up to 10 and differencing orders of 0 or 1. Candidate models were evaluated using AIC, BIC, and HQIC selection criteria. All three criteria consistently selected an ARIMA(1,1,1), consistent with the ACF and PACF.

The selected ARIMA(1,1,1) model can be written as:

$$\Delta y_t = \phi_1 \Delta y_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1}, \tag{1}$$

Where

$$y_t = \ln\left(\frac{CL2_t}{CL1_t}\right), \qquad \Delta y_t = y_t - y_{t-1}. \tag{2}$$

The ARIMA(1,1,1) model was estimated on 248 monthly training observations. Both the autoregressive and moving average coefficients are highly statistically significant, indicating strong short-run dependence in first differences of the log spread. The significant AR and MA components suggest that while the series exhibits persistence, short-term shocks are partially corrected over subsequent periods. This specification serves as the baseline benchmark against which the ARIMAX model and trading performance are evaluated.

4.3 ARIMAX Model

The ARIMAX(2,1,0) model is defined as:

$$\Delta y_t = \beta_1 CPI_{t-1} + \beta_2 Yield_{t-1} + \beta_3 SPX_{t-1} + \beta_4 DXY_{t-1} + \beta_5 VIX_{t-1} + \phi_1 \Delta y_{t-1} + \phi_2 \Delta y_{t-2} + \varepsilon_t \tag{3}$$

Where

$$y_t = \ln\left(\frac{CL2_t}{CL1_t}\right), \qquad \Delta y_t = y_t - y_{t-1}.$$

The ARIMAX specification incorporates lagged macroeconomic and market variables. Aligning these predictors with the spread and removing missing values reduced the training sample from 355 to 54 observations, an 85% reduction reflecting non-overlapping date ranges and the one-month lag to prevent look-ahead bias.After correlation filtering (threshold 0.85) and VIF filtering (threshold 10), five variables remained: CPI YOY, 3-month yield, S&P 500, US Dollar returns, and VIX. Grid search evaluated specifications with p,q < 4.

The ARIMAX(2,1,0) model was selected based on information criteria (shown on Table 6), achieving an AIC of minus 363.65 using 54 observations. Among the exogenous variables, only SPX and VIX coefficients are statistically significant at the 5% level. Both coefficients are negative, suggesting that stronger equity markets and higher volatility are associated with narrower futures log spreads. The remaining macro variables and AR terms are not statistically significant.
However it's important to note that the severe sample reduction raises critical concerns. With 54 observations supporting 9 parameters, the model violates guidelines of 10-15 observations per parameter, substantially limiting reliable estimation and likely explaining poor out-of-sample performance examined in Section 7.

For completeness, full coefficient estimates for the ARIMA and ARIMAX models are reported in Tables 8 and 9; the analysis focuses on forecast performance rather than coefficient-level inference.

5 Model Diagnostics (Residual Evaluation)

In this section, we will evaluate the models side-by-side to see whether each one has captured time dependence, if there has been any systematic information left in the error term, and whether our models are statistically stable enough to trust for forecasting. Our diagnostic analysis will primarily focus on the behaviour of model residuals as well-behaved and stable residuals are necessary for assessing model adequacy in time-series forecasting. In particular, residual diagnostics are used to evaluate whether each specification captures enough of the underlying dynamics of oil prices. This assessment is key to determining reliability of resulting forecasts rather than coefficient-level inference.

When assessing both of the models' residuals, we see that there are noticeable differences between the two specifications. While the ARIMA model produces residuals that are broadly centred around zero, the residual series exhibits periods of elevates volatility and several large shocks, suggesting that some systematic variation remains unexplained, particularly during turbulent market conditions. The residual distribution also displays mild heavy-tailed behaviour, indicating deviations from normality.

In contrast, the ARIMAX model yields residuals with reduced dispersion and more stable behaviour over time. The inclusion of exogenous variables appears to absorb part of the variation previously left in the residuals, resulting in fewer extreme observations and a tighter residual distribution. Although residuals remain imperfectly Gaussian, the ARIMAX specification provides a cleaner fit, supporting the use of additional economic information in modelling oil futures spreads.

To confirm findings, formal diagnostic tests were conducted to assess whether there is enough statistical evidence to suggest the residuals exhibit white noise behaviour. In theory, for an appropriate forecasting model to be utilised, it should capture all systematic time-series patterns, leaving only unpredictable noise in the residuals. For a useful forecasting model, residuals should exhibit no remaining serial dependence, while still reflecting the volatility and tail behaviour typical of oil markets. An insignificant Ljung–Box test indicates that the model has captured the available time-series predictability, leaving no exploitable autocorrelation. In contrast, it is common to reject the null hypotheses of homoskedasticity (ARCH) and normality (Jarque–Bera), as oil prices are characterised by volatility clustering and heavy-tailed shocks. Together, these diagnostics suggest a model that adequately forecasts the conditional mean while retaining realistic risk characteristics.

Table 1 highlights clear differences in residual behaviour between the ARIMA and ARIMAX specifications. For both models, the Ljung–Box test fails to reject the null hypothesis of no residual autocorrelation, indicating that serial dependence has been adequately captured and that no obvious linear predictability remains unexploited. However, the ARCH LM test strongly rejects homoskedasticity for the ARIMA model, suggesting the presence of volatility clustering in the residuals. In contrast, the ARIMAX model yields an insignificant ARCH LM p-value, indicating a substantial reduction in residual heteroskedasticity following the inclusion of exogenous variables. Both ARIMA and ARIMAX residuals strongly reject normality according to the Jarque–Bera test, indicating the presence of heavy tails and non-Gaussian shocks typical in oil markets. Overall, while ARIMAX results suggest that incorporating macroeconomic and market variables improves model adequacy by absorbing variation that would otherwise manifest as residual volatility and non-normal shocks, while maintaining satisfactory forecasting properties, this improvement does not translate into superior out-of-sample forecasting performance

Table 1: Residual diagnostic tests (p-values)

Test	**ARIMA(1,1,1)**	**ARIMAX(2,1,0)**
Ljung–Box (lag 10)	0.658	0.959
ARCH LM (lag 10)	0.022	0.734
Jarque–Bera	0.000	0.000

6 Out-of-Sample (OOS) Performance

6.1 Non-Rolling Forecasts

Figure 4 compares non-rolling out-of-sample forecasts from ARIMA and ARIMAX models against the realised log spread. Visually, the ARIMAX forecasts exhibit greater variation than the relatively flat ARIMA forecast, which may initially suggest superior responsiveness to changes in the series. However, the out-of-sample performance metrics reported in Table 2 indicate that this visual impression is misleading. In particular, the ARIMA(1,1,1) model achieves a substantially lower RMSE and MAE than the ARIMAX specification, as well as higher out-of-sample R^2 relative to both the last-observation and historical-mean benchmarks. In contrast, the ARIMAX model exhibits large forecast errors and strongly negative out-of-sample R^2, implying underperformance relative to simple benchmark forecasts. Overall, results suggest additional exogenous regressors in the ARIMAX model do not improve predictive accuracy, and that the more parsimonious ARIMA specification delivers superior average out-of-sample forecast performance.

Figure 4: Non Rolling OOS Forecasts (ARIMA vs ARIMAX)

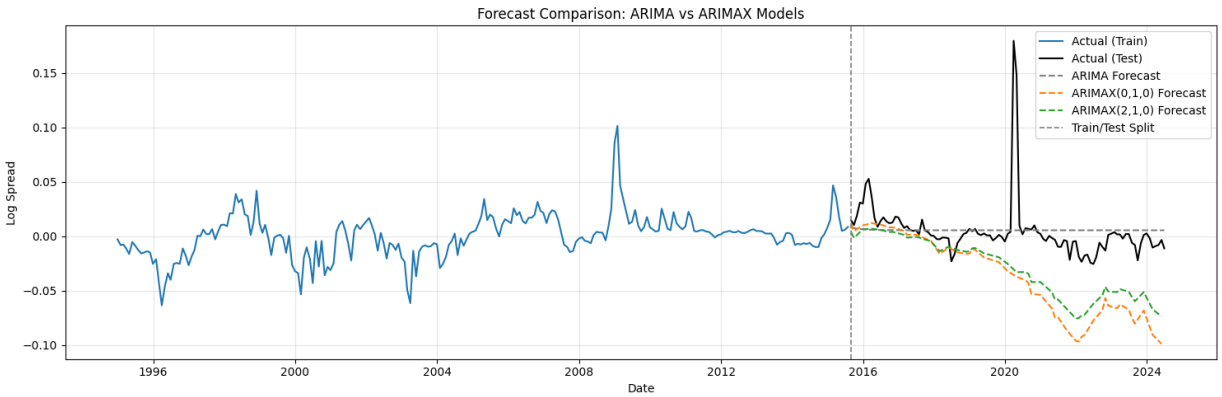


Table 2: Out-of-sample forecasting performance

Model	Hitrate	MAE	RMSE	OOS R^2 vs LastObs	OOS R^2 vs TrainMean
Benchmark_LastObs	0.0000	0.01143	0.02998	0.0000	-0.5539
Benchmark_TrainMean	0.5714	0.01138	0.02405	0.3565	0.0000
ARIMA(1,1,1)	0.5234	0.0128	0.0256	-0.2520	0.0254
ARIMAX(2,1,0)	0.5510	0.03537	0.04582	-1.3359	-2.6298

LastObs = Last Observation (Random Walk), TrainMean = (Historical Mean Forecast)

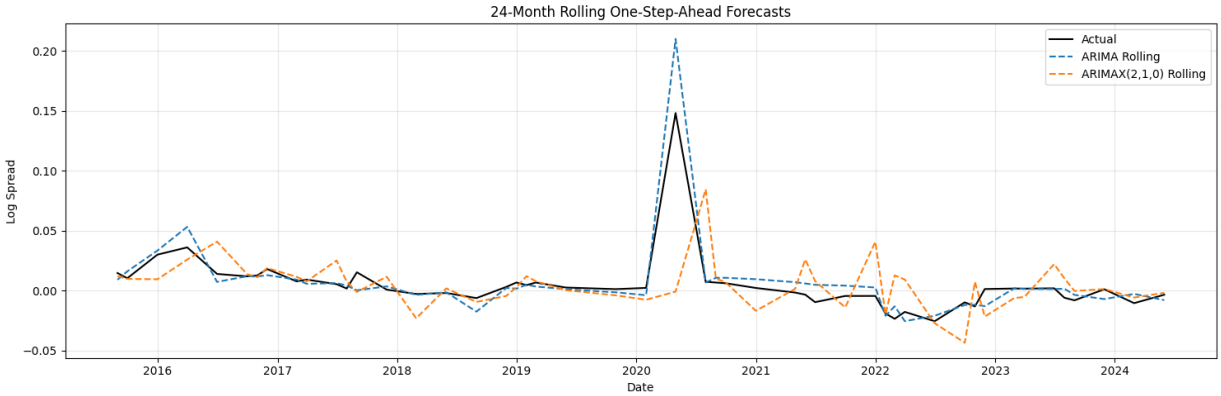
Although the non-rolling forecasts offer a clear comparison across models, they assume a static estimation sample. In practice, forecasts are updated as new information arrives. To reflect this, we adopt a rolling window forecasting approach in which model parameters are re-estimated over a 24-month window, allowing forecasts to better capture changing dynamics in the underlying series.

6.2 Rolling Forecasts

Figure 5 presents 24-month rolling window one-step-ahead out-of-sample forecasts for the ARIMA and ARIMAX models. The ARIMA specification closely tracks the realised series and captures the general pattern of the out-of-sample data, whereas the ARIMAX forecasts exhibit greater volatility and frequent deviations from the realised values. This suggests that the ARIMA model delivers more stable and accurate forecasts in a rolling setting, while the ARIMAX model may be adversely affected by additional noise introduced through the exogenous regressors. This visual evidence is consistent with the out-of-sample error metrics, which show lower forecast errors for ARIMA, reinforcing the conclusion that increased model complexity does not translate into improved predictive performance in a rolling window framework.

Additionally, reinforcing the earlier discussion on sample-size limitations, it is important to note that when fitting the ARIMAX model in the rolling window framework, only 54 observations are available within each window. This further increases estimation uncertainty and helps explain the weaker and noisier out-of-sample forecasts observed in this section.

Figure 5: 24-Month Rolling Window OOS Forecasts (ARIMA vs ARIMAX)



7 Trading Strategy

To provide an economic interpretation of the statistical forecasts, we translate the models' predictions (from above) of the CL2–CL1 calendar spread into a simple trading rule. Using the one-step-ahead forecast, the trading decision is defined as follows:

- Take a long position in the CL2–CL1 spread when the forecasted spread exceeds the most recent observed spread.
- Take a short position in the spread otherwise.

This rule is intended as a stylised illustration of how forecast information could be used by an oil trader, rather than as a fully specified trading strategy. It abstracts from transaction costs, position sizing, and risk management considerations, and serves to link the econometric forecasting exercise to an economically meaningful decision framework.

$$\text{Signal}_{t+1} = \begin{cases} +1 & \text{if } \hat{S}_{t+1} > S_t, \\ -1 & \text{otherwise.} \end{cases} \tag{4}$$

Where $S_t = \text{CL2}_t - \text{CL1}_t$ denotes observed spread, and $\hat{S}_{t+1}$ denotes one-step-ahead model-based forecast of spread.

7.1 Cumulative Economic Returns

Cumulative economic returns show how a trading strategy actually makes or loses money over time, taking into account the full sequence of trades rather than isolated outcomes. For an oil trader, this provides a clear picture of whether a model consistently adds value or whether profits are driven by a small number of lucky periods. A steadily rising cumulative return indicates a robust and repeatable signal that survives changing market conditions, while flat or declining cumulative performance suggests that the strategy is either too noisy, overfit, or eroded by transaction costs. As such, cumulative returns are a practical way to assess whether a forecasting model translates into economically meaningful and tradeable performance.

Looking at Figure 6, we see the ARIMA model performing significantly better than ARIMAX, particularly during the COVID crisis. This suggests that the simpler, price-driven dynamics captured by ARIMA were more robust to the

extreme volatility and structural breaks observed during this period, while the ARIMAX model struggled to extract stable signals from the exogenous variables, which in practice appear to have introduced noise rather than improving predictive power. Furthermore, Table 3 provides additional support for the rolling window results, with the ARIMA model achieving a higher directional hit rate of 63% alongside a markedly lower RMSE, suggesting more reliable and accurate forecasts than the ARIMAX model.

Figure 6: Cumulative Economic Returns

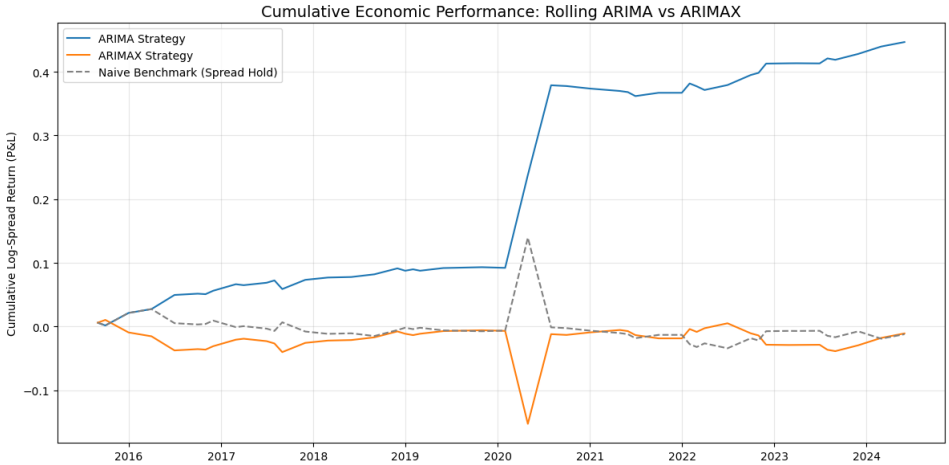


Table 3: Rolling Window Forecast Accuracy

Model	MAE	RMSE	Hitrate
ARIMA (Rolling)	0.006236	0.011009	0.632653
ARIMAX(2,1,0) (Rolling)	0.015300	0.028437	0.510204

7.2 PnL Analysis

Table 4 reports the performance of the rolling window trading strategies. The ARIMA-based strategy delivers the strongest economic performance across all key metrics, achieving the highest total and average PnL alongside an annualised Sharpe ratio above one, indicating attractive risk-adjusted returns. In addition, ARIMA exhibits the smallest maximum drawdown, suggesting greater robustness during adverse market conditions, while maintaining moderate turnover relative to the ARIMAX strategy.

By contrast, the ARIMAX-based strategy generates negligible profitability, a very low Sharpe ratio, and a substantially larger drawdown despite significantly higher turnover, implying that the added model complexity and trading activity do not translate into improved economic performance. In practice, transaction costs would likely reduce the profitability of higher-turnover strategies, particularly for the ARIMAX model.

Overall, these results highlight the effectiveness of the ARIMA model in producing stable, tradeable signals and reinforce the conclusion that simpler, price-based dynamics outperform more complex specifications in a rolling window trading framework.

Table 4: rolling window Trading Strategy Performance

Strategy	Total PnL	Avg. PnL	Sharpe (Ann.)	Max Drawdown	Turnover
ARIMA Rolling	0.446680	0.009116	1.094421	-0.016995	40.0
Benchmark TrainMean	0.371585	0.004764	0.691818	-0.027738	26.0
ARIMAX(2,1,0) Rolling	0.041071	0.000527	0.074985	-0.170758	76.0

8 Conclusion

This study set out to evaluate whether econometric time-series models can be used to forecast the WTI crude oil futures spread and whether such forecasts can be translated into economically meaningful trading strategies. Across both non-rolling and rolling out-of-sample evaluations, the ARIMA model consistently outperformed the ARIMAX specification in terms of forecast accuracy, directional hit rate, and economic performance. While the inclusion of macroeconomic and market variables in the ARIMAX model improved certain in-sample diagnostics, these gains did not translate into superior out-of-sample forecasts, particularly in volatile market regimes such as the COVID period.

The rolling window analysis, which better reflects real-world forecasting practice, further reinforced these findings. ARIMA produced more stable forecasts, lower forecast errors, and higher directional accuracy, which translated into a trading strategy with superior cumulative returns, higher risk-adjusted performance, and smaller drawdowns relative to both the ARIMAX strategy and benchmark alternatives. In contrast, the ARIMAX-based strategy generated negligible profitability and suffered from high turnover and large drawdowns, suggesting that the additional model complexity introduced noise rather than improving tradeable signal quality.

Importantly, the results suggest that the underperformance of ARIMAX may be driven, at least in part, by data limitations. Aligning exogenous variables and avoiding look-ahead bias reduced the effective sample size to just 54 observations within each rolling window, severely constraining reliable parameter estimation. Given these constraints, the use of more complex machine learning models such as Random Forests or LSTMs was not pursued, as such approaches typically require substantially larger datasets to avoid over fitting and to deliver robust out-of-sample performance. In this context, parsimonious econometric models offer a more appropriate balance between flexibility, interpretability, and robustness.

To summarise, the results demonstrate that simple, price-driven dynamics captured by ARIMA models are sufficient to extract economically meaningful information from oil futures spreads. The findings highlight an important practical insight for energy markets: in environments characterised by structural breaks, regime shifts, and limited data availability, simpler models can outperform more complex alternatives by delivering more robust and tradeable forecasts.

9 Appendix

9.1 Figures

Figure 7: ACF and PACF of Log Monthly Spread

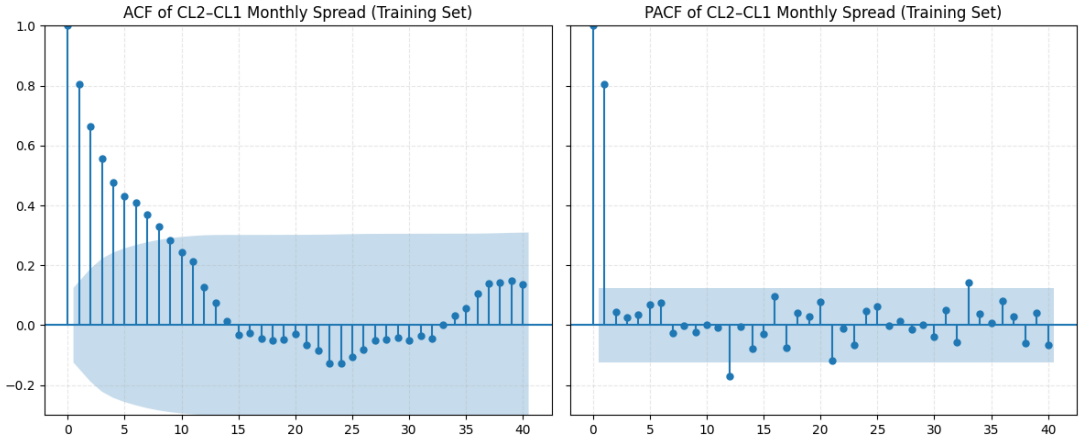


Figure 8: Residual plots of ARIMA(1,1,1)

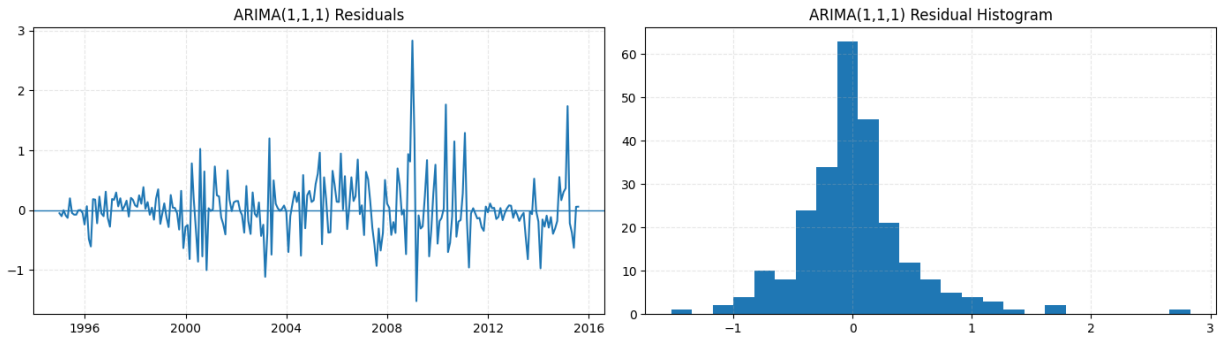


Figure 9: ARIMA Residuals Autocorrelation Plot

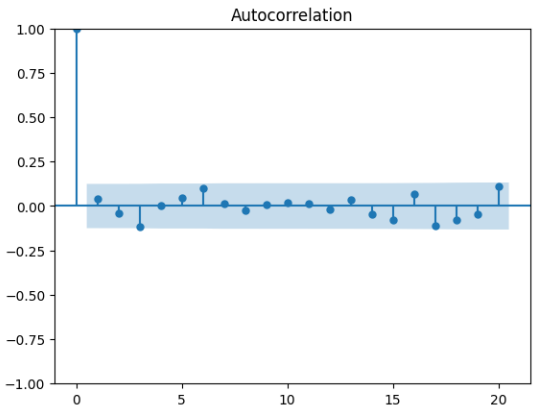


Figure 10: ARIMA(1,1,1) OOS Forecast Plot

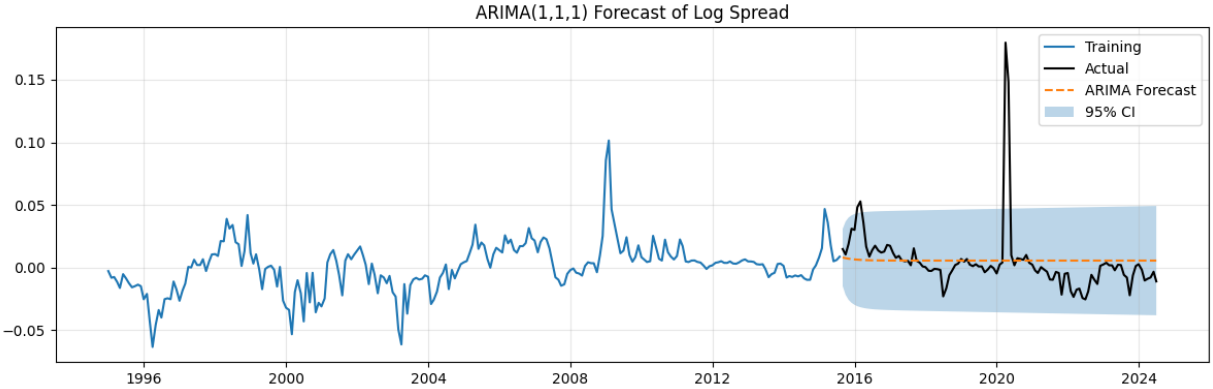
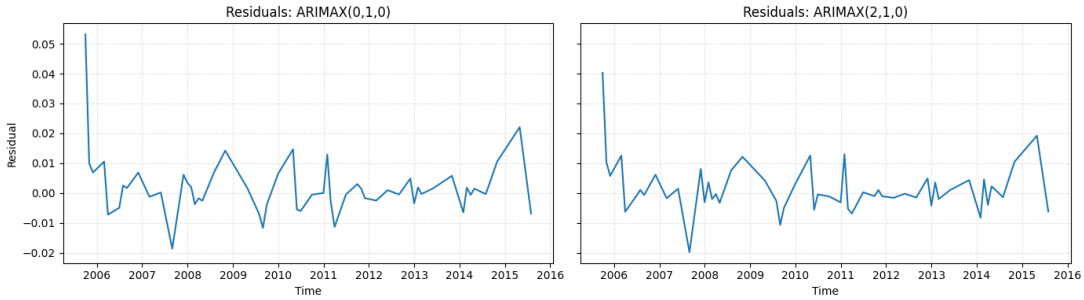
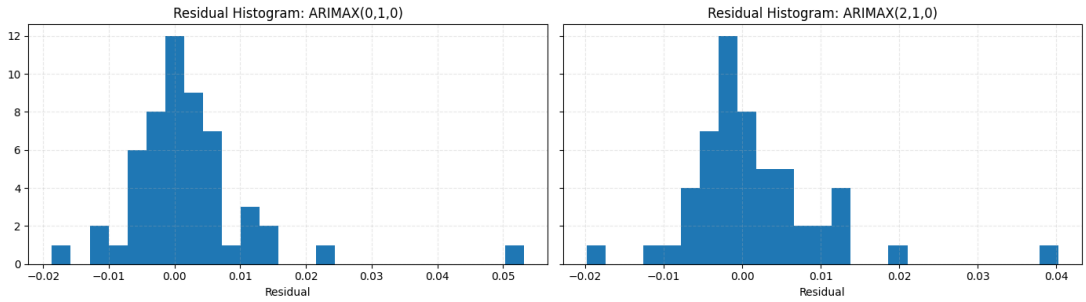


Figure 11: Residual diagnostics for ARIMAX models

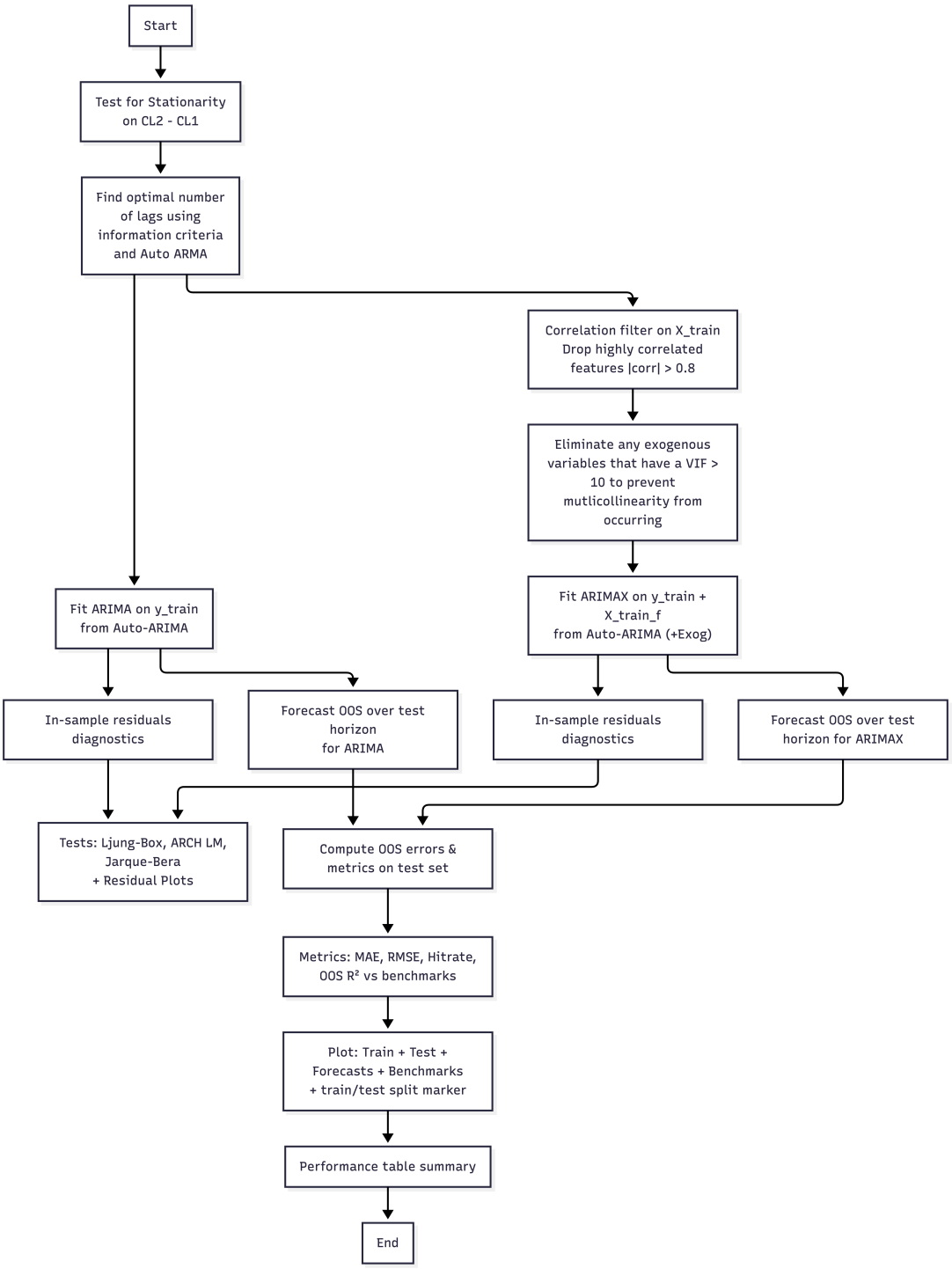


(a) Residual time-series plots for ARIMAX(0,1,0) and ARIMAX(2,1,0)



(b) Residual histograms for ARIMAX(0,1,0) and ARIMAX(2,1,0)

Figure 12: Methodology Overview



9.2 Tables

Table 5: Stationarity Tests for CL2–CL1 Spread Variants

Spread Specification	**ADF p-value**	**KPSS p-value**
CL2–CL1 Spread	0.0000	0.0177
CL2–CL1 Spread (Differenced)	0.0000	0.1000
log(CL2–CL1) Spread	0.0000	0.0230

Note: Here, we use a 99% Confidence Interval (1% α)

Table 6: Information criteria for alternative ARIMAX specifications

ARIMAX(p, d, q)	**AIC**	**BIC**	**HQIC**
(0,1,0)	−355.89	−344.18	−351.40
(2,1,0)	−350.27	−334.81	−344.36
(2,1,2)	−348.31	−329.19	−341.03
(0,1,2)	−345.88	−330.59	−340.06
(2,1,1)	−338.58	−321.20	−331.94
(3,1,3)	−337.72	−315.02	−329.11
(4,1,3)	−335.31	−310.71	−325.98

Table 7: Model selection criteria and diagnostic tests for ARIMAX specifications

Model	**Order**	**AIC**	**BIC**	**HQIC**	**Ljung–Box p**	**ARCH p**	**JB p**
ARIMAX	(0,1,0)	−361.75	−349.93	−357.20	0.844	0.884	4.15×10^{-74}
ARIMAX	(2,1,0)	−363.65	−347.89	−357.59	0.959	0.734	6.68×10^{-27}

Table 8: (ARIMA(1,1,1)) Results

	ARIMA(1,1,1)
$ar(1)$	0.7826***
	(0.0350)
$ma(1)$	-0.9767***
	(0.0170)
σ^2	0.2318***
	(0.0110)
Observations	248
Log Likelihood	-170.518
AIC	347.037
BIC	357.565
HQIC	351.275
Ljung–Box $Q(1)$	0.42
Prob(Q)	0.52
Jarque–Bera (JB)	365.34
Prob(JB)	0.00
Heteroskedasticity (H)	3.33
Prob(H)	0.00
Skewness	1.14
Kurtosis	8.50

Notes: Standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 9: ARIMAX Model Estimates for ln(spread)

	ARIMAX(0,1,0)	ARIMAX(2,1,0)
CPI YoY (%)	-0.0002	-0.0005
	(0.0010)	(0.0010)
3M Yield	0.0012	0.0012
	(0.0030)	(0.0030)
SPX	-3.876e-05*	-2.834e-05**
	(2.11e-05)	(1.43e-05)
DXY returns	0.0097	0.0097
	(0.0340)	(0.0300)
VIX	-0.0007***	-0.0006**
	(0.0000)	(0.0000)
$ar(1)$	-	0.1158
		(0.1660)
$ar(2)$	-	-0.3319
		(0.2250)
σ^2	5.074e-05***	4.452e-05***
	(9.21e-06)	(9.29e-06)
Observations	54	54
Log Likelihood	186.874	189.825
AIC	-361.748	-363.651
BIC	-349.927	-347.889
HQIC	-357.202	-357.590
Ljung–Box $Q(1)$	0.38	0.59
Prob(Q)	0.54	0.44
Jarque–Bera	3.13	2.91
Prob(JB)	0.21	0.23
Heteroskedasticity (H)	0.90	0.78
Prob(H)	0.83	0.60
Skewness	0.27	0.14
Kurtosis	4.06	4.11

Notes: Standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 10: Rolling window out-of-sample forecast performance

Model	**MAE**	**RMSE**	**Hit Rate**
ARIMA (Rolling)	0.0062	0.0110	0.633
ARIMAX(2,1,0) (Rolling)	0.0153	0.0284	0.510